30 Days SQL Micro Course Certificate Assignment

1. Total Number of Patients

Find the total number of patients across all hospitals.

Answer:-

```
select sum(patients_count) as total_patients from hospitals;
```

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2. Average Number of Doctors per Hospital

Retrieve the average count of doctors available in each hospital.

Answer:-

```
select hospital_name, avg(doctors_count) as avg_doctors from hospitals group by hospital_name;
```

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3. Top 3 Departments with the Highest Number of Patients

Find the top 3 hospital departments that have the highest number of patients.

Answer:-

```
select department, sum(patients_count) as total_patients from hospitals group by department order
by total_patients desc limit 3;
```

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4. Hospital with the Maximum Medical Expenses

Identify the hospital that recorded the highest medical expenses.

Answer:-

```
select distinct hospital_name, max(medical_expenses) as max_expenses from hospitals group by
hospital_name order by max_expenses desc limit 1;
```

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5. Daily Average Medical Expenses

Calculate the average medical expenses per day for each hospital.

Answer:-

```
select hospital_name,avg(medical_expenses) as avg_daily_expenses from hospitals
group by hospital_name;
```

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6. Longest Hospital Stay

Find the patient with the longest stay by calculating the difference between Discharge date and Admission date

Answer:-

```
select hospital_name,department,(discharge_date - admission_date) as
stay_duration from hospitals order by stay_duration desc limit 1;
```

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7. Total Patients Treated Per City

Count the total number of patients treated in each city.

Answer:-

```
select location,sum(patients_count) as total_patients from hospitals group by
location;
```

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8. Average Length of Stay Per Department

Calculate the average number of days patients spend in each department.

Answer:-

```
select department, avg(discharge_date - admission_date) as avg_stay_days from hospitals group by
department;
```

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9. Identify the Department with the Lowest Number of Patients

Find the department with the least number of patients.

Answer:-

```
select department ,sum(patients_count) as total_patients from hospitals group by department order
by total_patients asc limit 1;
```

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10. Monthly Medical Expenses Report

Group the data by month and calculate the total medical expenses for each month.

Answer:-

```
select extract(month from admission_date) as month, sum(medical_expenses) as
total_expenses from hospitals group by month order by month;
```
